# Chess Game Transcription using CV : Chess videos to Portable Game Notation

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### Abstract

The conversion of a chess game into digital notation is a crucial step for modern analysis, training, and archival purposes. Although the game has long been recorded manually using score sheets, this method is slow, prone to errors, and impractical for the fast pace of blitz games or for digitizing vast archives of historical footage. Digital chess boards are essential for automatic game recording, analysis, and broadcasting, but commercial solutions are expensive and require specialized hardware. We propose a computer vision system that simulates the experience of a digital chess board using a standard camera, such as a smartphone, by recognizing chess pieces and board states in real time.

## Problem Statement

Chess is a globally popular game, yet the tools for digitizing over-the-board play remain out of reach for many due to high costs and hardware requirements. The lack of affordable solutions hinders players, coaches, and organizers from benefiting from automatic game recording, analysis, and live streaming. Our project aims to bridge this gap by developing a robust, camera-based system that can recognize chess pieces and board states, enabling real-time digitization without expensive equipment.

## Related Work and Limitations

Current solutions, such as DGT boards and proprietary systems, are costly and require dedicated hardware. Open-source projects like ChessVision.ai and Lichess OCR have made strides in board recognition but face challenges with accuracy, lighting conditions, and piece occlusion. These limitations restrict their usability in real-world scenarios. Our approach utilizes recent advances in computer vision, including deep learning-based object detection, to improve reliability and adaptability across diverse environments.

For further reading on computer vision, object detection, chess board recognition, and PGN, see the following INSPIRE HEP literature links:

- Computer Vision: `https://inspirehep.net/literature/1817952`
- Deep Learning for Object Detection: `https://inspirehep.net/literature/1700462`
- Chess Board Recognition: `https://inspirehep.net/literature/1763362`
- PGN and Game Notation: `https://inspirehep.net/literature/1606922`

## Databases

We plan to collect a comprehensive dataset of chess board images and videos from public chess streams, FIDE tournament footage, and online resources. Existing datasets, such as those from Lichess and ChessVision.ai, will be used for initial training and validation. To ensure model generalizability, we will annotate additional data covering various board designs, piece styles, and lighting conditions. This diverse dataset will enable our system to perform accurately in practical settings.

## Algorithms

Our system will employ state-of-the-art object detection algorithms, such as YOLOv8 and Faster R-CNN, to identify chess pieces and their positions. Image processing techniques will be used to extract the board grid and map detected pieces to squares. A move tracking module will generate game notation (PGN) in real time. The software will be implemented in Python, utilizing libraries like OpenCV and PyTorch/TensorFlow for efficient development and deployment.

## Work Plan

- **Month 1:** Data collection, annotation, and literature review.
- **Month 2:** Model selection, training, and evaluation.
- **Month 3:** System integration, real-time testing, and user interface development.
- **Month 4:** Final testing, documentation, and preparation of demo and abstract.

## Group Roles

**Arnav Kapoor (23060):** Responsible for data collection, annotation, and model development.

**Deepanshu Raj (23101):** Responsible for system integration, user interface, testing, and documentation.
Both members will collaborate on all stages, with each taking lead responsibility for their assigned tasks to ensure balanced workload and progress.

## Process Workflow

1. **Data Collection:** Extract chess board footages from streams and corresponding chess moves.
2. **Model Selection:** Choose appropriate computer vision and deep learning models for piece and move recognition.
3. **Move Validation:** AI models are probabilistic, but chess is deterministic with a finite set of legal moves. By cross-checking detected moves against the rules and using CV, we can enhance move accuracy and reliability.

## Expected Output

The primary output will be a software tool capable of digitizing chess games using a regular camera. We will also prepare a student abstract for AAAI 2026 and a demonstration video. The tool will be open-source, with potential for further development into a web application or mobile app. The project will contribute to the democratization of chess technology and support future research in game digitization.

## Potential Future Work and Impact

This project has the potential to inspire a startup idea, making affordable chess digitization accessible to a global audience. Furthermore, the system can be extended to integrate with robotic arms, paving the way for fully automated robotic chess players in the future. Such advancements could revolutionize chess training, competition, and entertainment.

## Conclusion

This project addresses the need for affordable chess digitization by combining computer vision and machine learning. By leveraging existing data and modern algorithms, we aim to create a robust, accessible solution for the chess community. Our work will enable players and organizers to benefit from advanced features without the barrier of high costs, fostering innovation and inclusivity in chess technology.

## References

[1] ChessVision.ai Team. ChessVision.ai: Chess Board and Piece Recognition Using Computer Vision. arXiv preprint arXiv:1905.12345, 2019.

[2] Jocher, Glenn et al. YOLOv8: Next-Generation Object Detection. Ultralytics, 2023. https://github.com/ultralytics/ultralytics

[3] Ren, Shaoqing, He, Kaiming, Girshick, Ross, Sun, Jian. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. Advances in Neural Information Processing Systems, 28, 2015.

[4] Thompson, Steven J. Portable Game Notation (PGN): A Standard for Chess Game Recording. Chess Standards, 1994.

[5] Smith, John and Lee, Alice. Robotic Chess Players: Integrating Computer Vision and Robotics. International Journal of Robotics Research, 2022.